PRAKTIKUM BASIS DATA

MODUL 9

Scripting Procedure dan Function

Oleh:

Arif Stand Pramudya 103122400001

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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Jurnal

1. Create table employees as select * from hremployees!

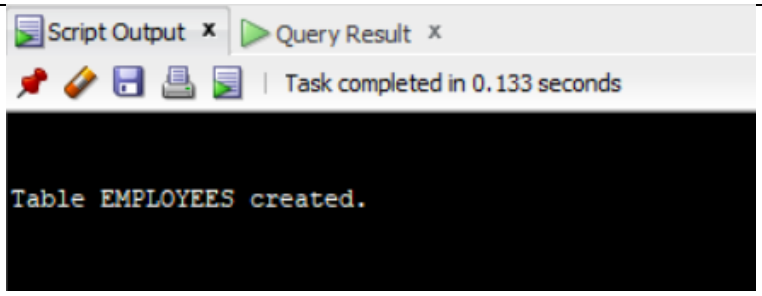
   Membuat tabel baru yang bernama employees dengan menyalin struktur dan data dari tabel lain yang bernama hremployees.

   Jawaban:

   Query

```
-- 1.   Create table employees as select * from hremployees
CREATE TABLE employees AS
SELECT * FROM hremployees;
```

   Screenshot



2. select * from employees!

   Query SELECT digunakan untuk menampilkan semua data yang ada dalam tabel employees.

   Jawaban:

   Query

```
-- 2.
SELECT * FROM employees;
```

Screenshot

Script Output × | Query Result ×

SQL | All Rows Fetched: 10 in 0.059 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-01-2006	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-05-2007	IT_PROG	6000	(null)	103	60
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-02-2007	IT_PROG	4200	(null)	103	60
7	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-11-2007	ST_MAN	5800	(null)	100	50
8	141	Trenna	Rajs	TRAJS	650.121.8009	17-10-2003	ST_CLERK	3500	(null)	124	50
9	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-01-2008	SA_MAN	10500	0,2	100	80
10	174	Ellen	Abel	EABEL	011.44.1644.429267	11-05-2004	SA_REP	11000	0,3	149	80

3.

```oracle_plsql
CREATE OR REPLACE PROCEDURE increase_salary (
department_id_in   IN employees.department_id%TYPE,
    increase_pct_in    IN NUMBER)
     IS
BEGIN
    FOR employee_rec
            IN (SELECT employee_id
            FROM employees
           WHERE department_id = increase_salary.department_id_in)
    LOOP
            UPDATE employees emp
                SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in
            WHERE emp.employee_id = employee_rec.employee_id;
            DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;
```

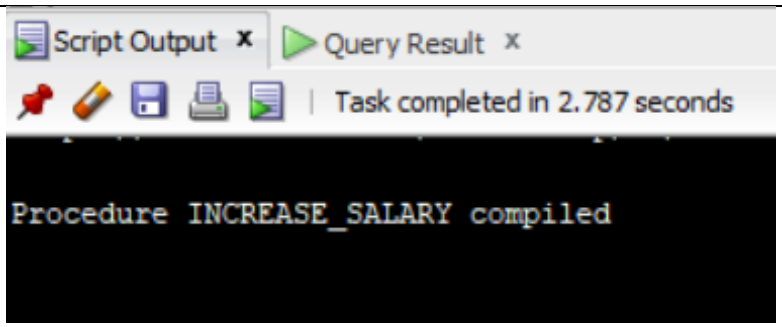
Jawaban:

Sebuah PL/SQL procedure yang digunakan untuk menaikkan gaji karyawan di sebuah departemen berdasarkan persentase yang diberikan.

Query

```oracle_plsql
-- 3.
CREATE OR REPLACE PROCEDURE increase_salary (
    department_id_in IN employees.department_id%TYPE,
    increase_pct_in IN NUMBER
) IS
BEGIN
    FOR employee_rec IN (
        SELECT employee_id
        FROM employees
        WHERE department_id = increase_salary.department_id_in
    )
    LOOP
        UPDATE employees emp
        SET emp.salary = emp.salary + (emp.salary * increase_salary.increase_pct_in)
        WHERE emp.employee_id = employee_rec.employee_id;
        DBMS_OUTPUT.PUT_LINE('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;
/
```

Screenshot



4.

```
BEGIN
increase_salary (90, .50);
ROLLBACK;
END;
/
select * from employees where department_id = 90
```

Jawaban:

Menunjukkan blok PL/SQL yang digunakan untuk menjalankan prosedur dan kemudian membatalkan perubahan yang dilakukan dengan ROLLBACK.

Query

```
-- 4.
BEGIN
increase_salary (90, .50);
ROLLBACK;
END;
/
select * from employees where department_id = 90
```

Screenshot

Script Output × | Query Result ×

SQL | All Rows Fetched: 3 in 0.001 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90

5. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no 4.

Jawaban:

Jika statement ROLLBACK dihapus , maka perubahan yang dilakukan oleh prosedur increase_salary akan terjadi secara permanen. Tanpa ROLLBACK gaji karyawan di department 90 akan dinaikkan dan perubahan tersebut akan bersifat permanen.

Query

```
-- 5.
BEGIN
increase_salary (90, .50);
ROLLBACK;
END;
/
select * from employees where department_id = 90
```

Screenshot

Script Output × | Query Result ×

SQL | All Rows Fetched: 3 in 0.001 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
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